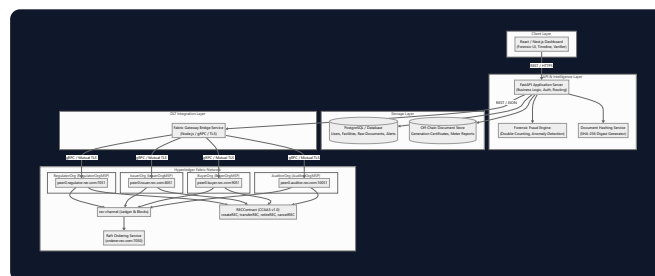


System Architecture

Renewable Energy Certificate (REC) Fraud Detection & Tracking System

This document outlines the end-to-end architecture of the REC Fraud Detection & Tracking System, integrating **Hyperledger Fabric v2.5** distributed ledger technology with an off-chain **FastAPI** backend, **PostgreSQL** relational database, and modern **React / Next.js** forensic intelligence frontend.

1. Architectural Overview



2. Why Permissioned Blockchain & Why Hyperledger Fabric?

2.1 Public vs Permissioned Blockchains

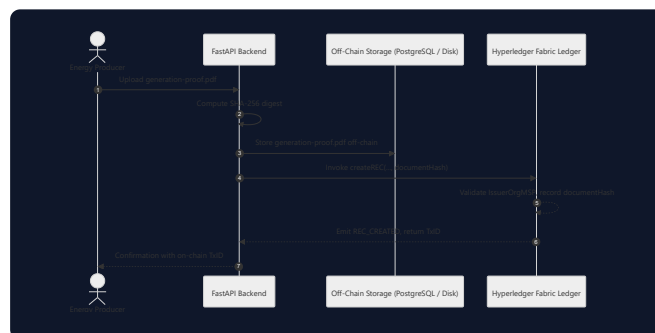
- **Public Blockchains (e.g. Ethereum, Solana):**
 - Publicly readable by anonymous entities, high and volatile transaction fees (gas), low throughput, and lack of native enterprise privacy partitions.
 - In regulatory and corporate REC trading, market participants must be strictly verified entities (Regulators, Certified Issuing Registries, Verified Corporate Buyers, Accredited Auditors).
- **Permissioned DLT (Hyperledger Fabric):**
 - **Known Identities:** Every transaction is cryptographically signed using X.509 PKI digital certificates issued by an authorized Certificate Authority.
 - **Deterministic Finality:** Uses Raft crash-fault-tolerant (CFT) ordering without probabilistic forks or proof-of-work/stake mining delays.
 - **Granular MSP Authorization:** Chaincode can directly inspect the caller's Membership Service Provider (MSP) identifier (e.g. `IssuerOrgMSP` vs `BuyerOrgMSP`) to enforce strict separation of duties.
 - **Zero Gas Fees:** High-throughput transaction settlement without transaction gas volatility.

2.2 Why Blockchain Instead of Only PostgreSQL?

Metric / Threat	PostgreSQL Only	Hyperledger Fabric DLT + PostgreSQL
Trust Model	Centralized database administrator has root access and can rewrite rows	Decentralized consensus across 4 independent organizations
Tamper-Evidence	Logs and records can be altered, truncated, or deleted	Cryptographically chained blocks; immutable audit history
Double Spending / Reuse	Depends on application-level locks vulnerable to DB replication race conditions	State machine transition enforced by multi-peer majority endorsement
Auditability	Difficult to prove external tamper-resistance to third-party auditors	Complete chronological transaction key history (<code>getHistoryForKey</code>)

3. On-Chain vs Off-Chain Data Separation

A foundational principle of enterprise blockchain architecture is **never store large binary documents on the ledger**.



What Goes On-Chain:

- Certificate identifier (`recId`)
- Generator facility ID (`generatorId`)
- Energy type (`energySource`)
- Generation period date (`generationDate`)
- Generation quantity (`generationMWh` , `issuedQuantity`)
- Current owner & Participant balance map (`balances: { [owner]: quantity }`)
- Active & Retired quantities (`activeQuantity` , `retiredQuantity`)
- Certificate lifecycle status (`ACTIVE` , `RETIRED` , `CANCELLED`)
- **Cryptographic SHA-256 document fingerprint** (`documentHash`)
- State change timestamps and Fabric Transaction IDs

What Stays Off-Chain:

- Raw PDF reports, meter readings, single-line diagrams, inverter logs
- User passwords, JWT sessions, profile details
- AI / ML model weights and training datasets

- Intermediate fraud scoring calculations and logs